

DOC_INTEL: Operations Guide

Daily Operations

Check Pipeline Health

```
SELECT * FROM DOC_INTEL.TELEMTRY.v_pipeline_health;
```

Green state: `pending_count = 0`, `failed_count = 0`, `processing_count = 0`

Action needed: `pending_count > 500` (check tasks), `failed_count > 0` (review errors)

Check for Errors

```
SELECT * FROM DOC_INTEL.TELEMTRY.v_error_summary
WHERE error_date = CURRENT_DATE()
ORDER BY occurrence_count DESC;
```

Monitor Cost

```
SELECT cost_date, SUM(total_credits) AS daily_credits
FROM DOC_INTEL.TELEMTRY.v_cost_by_folder
WHERE cost_date >= DATEADD('DAY', -7, CURRENT_DATE())
GROUP BY 1 ORDER BY 1 DESC;
```

Weekly Operations

Review Low-Confidence Classifications

```
SELECT file_path, source_folder, primary_document_type,
       confidence_score
FROM DOC_INTEL.DATA.classified_documents
WHERE needs_review = TRUE
```

```
AND classified_at >= DATEADD('DAY', -7, CURRENT_TIMESTAMP())
ORDER BY confidence_score ASC;
```

Review Pending Overrides

```
SELECT * FROM DOC_INTEL.ADMIN.extraction_overrides
WHERE override_status = 'PENDING'
ORDER BY submitted_at ASC;
```

Approve via admin UI or: `CALL`

```
DOC_INTEL.ADMIN.sp_approve_override('<override_id>');
```

Check Search Quality

```
SELECT query_date, tool_used, query_count, avg_latency_ms,
       positive_feedback_pct
FROM DOC_INTEL.TELEMTRY.v_search_analytics
WHERE query_date >= DATEADD('DAY', -7, CURRENT_DATE())
ORDER BY 1 DESC;
```

Pipeline Management

Force-Reprocess a Single Document

```
CALL DOC_INTEL.DATA.sp_force_reprocess(
    'documents/my-document.pdf',
    'Content changed on source system'
);
```

Bulk Reprocess a Source Folder

```
-- Dry run first
CALL DOC_INTEL.DATA.sp_bulk_reprocess('Claims', NULL, TRUE);

-- Execute if dry run looks correct
CALL DOC_INTEL.DATA.sp_bulk_reprocess('Claims', NULL, FALSE);
```

Suspend/Resume Tasks

```
-- Suspend (maintenance window)
ALTER TASK DOC_INTEL.DATA.task_01_register_files SUSPEND;
```

```
-- Resume
ALTER TASK DOC_INTEL.DATA.task_01_register_files RESUME;
```

Reset Stuck PROCESSING Documents

```
-- If tasks died mid-run, PROCESSING documents may be stuck
CALL DOC_INTEL.ADMIN.sp_reset_pipeline('CONFIRM_RESET');
```

Common Troubleshooting

Documents Not Appearing in Search

1. Check `ingestion_status` in registry:

```
SELECT ingestion_status, COUNT(*) FROM
DOC_INTEL.DATA.ingestion_registry GROUP BY 1;
```

2. If PENDING: tasks may be suspended. Check
`SHOW TASKS IN SCHEMA DOC_INTEL.DATA;`
3. If COMPLETE but not in catalog: Dynamic Table may not have refreshed. Check `SHOW DYNAMIC TABLES;`
4. If in catalog but not in search: Cortex Search may still be indexing. Wait for `TARGET_LAG`.

High Error Rate

1. Check error details:

```
SELECT last_error, error_stage, COUNT(*) FROM
DOC_INTEL.DATA.ingestion_registry
WHERE ingestion_status = 'FAILED' GROUP BY 1,2 ORDER BY 3
DESC;
```

2. Common causes:

- File too large (>100MB): Move to separate processing queue
- Unsupported format: Check file extension is in allowed list
- OCR failure: File may be corrupt or encrypted
- Network timeout: Retry usually resolves

Cortex Search Not Returning Results

1. Check search service status:

```
SHOW CORTEX SEARCH SERVICES IN SCHEMA DOC_INTEL.SERVICES;
```

2. Verify document_catalog has data:

```
SELECT COUNT(*) FROM DOC_INTEL.DATA.document_catalog;
```

3. Test with SEARCH_PREVIEW:

```
SELECT PARSE_JSON(SNOWFLAKE.CORTEX.SEARCH_PREVIEW(  
    'DOC_INTEL.SERVICES.doc_search_svc',  
    '{"query":"test","columns":  
        ["document_type"],"limit":3}'  
))[ 'results' ];
```

Adding a New User

```
USE ROLE SECURITYADMIN;  
  
-- Grant based on job function  
-- Standard user (search, browse, Q&A)  
GRANT ROLE DOC_INTEL_USER TO USER new.user@company.com;  
  
-- Pipeline operator (reprocess, manage tasks)  
GRANT ROLE DOC_INTEL_PIPELINE TO USER  
    pipeline.user@company.com;  
  
-- Admin (full access including config and overrides)  
GRANT ROLE DOC_INTEL_ADMIN TO USER admin@company.com;
```

Then share the app URL with the user. Retrieve it with:

```
snow app open --print-only
```

Updating the Classification Taxonomy

Classification labels and categories are now managed through the Admin UI:

1. Navigate to **Admin** → **Classification Labels**
2. **Add a label:** Enter a name, select (or create) a category, click Add
3. **Create a category:** Click the folder+ icon next to the category dropdown
4. **Rename a category:** Click the pencil icon on the category header
5. **Delete a category:** Click the trash icon on the category header (removes all labels in it)
6. **Disable a label:** Toggle the switch off (label won't be used for new classifications)

Changes take effect immediately for new documents. To reclassify existing documents, use the Reprocess tab on Pipeline Management.

SQL alternative:

```
-- Add a new label directly
INSERT INTO DOC_INTEL.ADMIN.classification_labels (label,
category, sort_order, is_active)
VALUES ('New Document Type', 'Claims', 100, TRUE);

-- Re-classify affected documents
CALL DOC_INTEL.DATA.sp_bulk_reprocess(NULL, NULL, FALSE);
```

Managing Lines of Business

Lines of Business are configurable through the Admin UI. The pipeline's LOB detection (AI_CLASSIFY) reads active LOBs from this table dynamically — no code changes needed to add or remove LOBs.

1. Navigate to **Admin** → **Lines of Business**
2. **Add a LOB:** Click “Add LOB”, enter code (e.g., `WC`), display name (e.g., `Workers Compensation`), optional description
3. **Edit a LOB:** Click the pencil icon to edit name/description inline
4. **Disable a LOB:** Toggle the switch off (LOB won't be available for new classifications)
5. **Delete a LOB:** Click the trash icon (use with caution — existing documents referencing this LOB code will retain their value)

Important: The `lob_code` must match what's used in `extraction_schemas.match_rule` for LOB-specific extraction to work. For example, if you add LOB code `Aviation`, you should also add extraction schema entries with `match_rule = '{"category":"...", "lob":"Aviation"}'`.

SQL alternative:

```
-- Add a new LOB
INSERT INTO DOC_INTEL.ADMIN.lines_of_business (lob_code,
lob_name, description, sort_order)
VALUES ('Aviation', 'Aviation Insurance', 'Aviation hull and
liability coverage', 11);

-- View active LOBs
SELECT lob_code, lob_name FROM
DOC_INTEL.ADMIN.lines_of_business
WHERE is_active = TRUE ORDER BY sort_order;
```

Backup & Recovery

The pipeline is fully reproducible from source files on the stage. To recover from data loss:

1. **Registry only:** Re-run `sp_register_new_files()` — rebuilds from stage
2. **Parsed documents:** Re-process via bulk reprocess
3. **Full wipe:** `DROP DATABASE DOC_INTEL CASCADE` then re-run all 7 SQL scripts
4. **Source documents:** Managed by ImageRight — no backup needed in Snowflake

Scaling Considerations

Dimension	Guidance
> 1M documents	Consider increasing <code>DOC_INTEL_WH</code> to LARGE during bulk initial load
High query volume	Create multiple <code>DOC_INTEL_QUERY_WH</code> warehouses for different user groups
Multiple regions	

Dimension	Guidance
	Set Cortex Search <code>CROSS_REGION = TRUE</code> if users span regions
Real-time requirements	Reduce <code>TARGET_LAG</code> on tasks and Cortex Search to 15 minutes